American Computer Science League

2025-2026 ● Contest 4: Numbrix ● Senior Division

PROBLEM: *Numbrix* is an *n* x *n* grid number puzzle. The objective is to fill the grid with all of the integers from 1 to n^2 inclusive by placing all missing integers, *x*, in the grid such that both *x - 1* and *x + 1* are adjacent to it, in the same row or the same column.

In this problem, you will be given the grid size, *n*, and the integers 1 through n^2 that fill the grid in row-major order. All missing numbers are represented by 0. Your task is to determine the missing numbers and output them in row-major order.

We guarantee that:

- There will be a unique solution to the given puzzle.
- The integers 1 and n^2 will never be one of the missing numbers.
- There will be no more than *2n* missing integers.

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EXAMPLE:

<table>
<tr><td>Input</td><td>5
3 0 1 0 23 0 11 0 25 22 5 10 13 20 0 6 0 14 19 18 0 8 0 16 0</td></tr>
<tr><td>Output</td><td>2 24 4 12 21 9 7 15 17</td></tr>
<tr><td>Explanation</td><td>The puzzle is represented in the 2-dimensional array as follows:
<table>
<tr><td>3</td><td>0</td><td>1</td><td>0</td><td>23</td></tr>
<tr><td>0</td><td>11</td><td>0</td><td>25</td><td>22</td></tr>
<tr><td>5</td><td>10</td><td>13</td><td>20</td><td>0</td></tr>
<tr><td>6</td><td>0</td><td>14</td><td>19</td><td>18</td></tr>
<tr><td>0</td><td>8</td><td>0</td><td>16</td><td>0</td></tr>
</table>
In order for the puzzle to be correct, the 0 spaces must be filled with the integers 2, 4, 7, 9, 12, 15, 17, 21, 24.
<table>
<tr><td>3</td><td>2</td><td>1</td><td>24</td><td>23</td></tr>
<tr><td>4</td><td>11</td><td>12</td><td>25</td><td>22</td></tr>
<tr><td>5</td><td>10</td><td>13</td><td>20</td><td>21</td></tr>
<tr><td>6</td><td>9</td><td>14</td><td>19</td><td>18</td></tr>
<tr><td>7</td><td>8</td><td>15</td><td>16</td><td>17</td></tr>
</table>
Output these in row-major order: 2 24 4 12 21 9 7 15 17</td></tr>
</table>

INPUT: Input an integer, *n*, for the size of the puzzle and a string containing all of the integers from 1 to n^2 in an *n* x *n* two-dimensional array, using a 0 for any integer that is missing. The value of *n* will be between 4 and 9, inclusive.

OUTPUT: Output a string of integers representing the missing number(s), in row-major order, each separated by a single space.

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SAMPLE INPUT:

1. 5
   3 0 1 0 23 0 11 0 25 22 5 10 13 20 0 6 0 14 19 18 0 8 0 16 0
2. 4
   7 8 0 0 6 1 12 11 5 0 0 0 0 3 16 15
3. 6
   28 0 26 23 0 21 0 30 0 24 19 20 0 31 36 1 0 17 33 0 35 0 15 16 0 5 0 3 14 13 7 8 0 10 11 0
4. 5
   11 0 25 24 23 12 9 8 0 0 0 0 1 6 21 14 3 4 5 0 0 16 0 18 0
5. 6
   27 28 0 30 31 32 0 25 0 17 0 0 1 24 0 16 0 36 2 23 20 15 14 13 3 0 21 8 9 12 4 5 6 0 10 11
6. 6
   16 0 18 0 26 0 15 0 19 24 29 28 14 21 22 0 0 31 0 2 3 4 5 32 12 1 0 7 6 33 0 10 9 36 0 34

SAMPLE OUTPUT:

1. 2 24 4 12 21 9 7 15 17
2. 9 10 2 13 14 4
3. 27 22 29 25 32 18 34 2 6 4 9 12
4. 10 7 22 13 2 20 15 17 19
5. 29 26 18 34 33 19 35 22 7
6. 17 25 27 20 23 30 13 8 11 35

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TEST DATA

TEST INPUT:

1. 7
   49 0 47 0 45 44 43 16 0 18 0 20 0 42 15 14 13 24 23 22 41 0
   1 0 0 0 33 40 3 10 11 26 31 34 0 0 9 8 27 0 35 38 5 6 0 28
   29 36 0
2. 7
   3 2 1 42 43 44 49 4 0 6 41 0 45 48 9 8 7 38 39 46 0 10 0 18
   0 36 35 34 11 16 19 20 21 32 33 0 15 24 23 22 31 0 13 14 0
   26 0 28 29
3. 8
   3 0 33 34 37 38 41 0 4 1 32 0 36 0 40 43 5 0 31 24 23 46 45
   0 8 7 30 0 22 0 52 53 9 0 29 26 21 48 51 0 12 11 28 0 20 49
   50 55 13 0 17 0 19 60 0 56 14 15 64 63 62 61 58 57
4. 8
   0 45 0 47 48 0 52 53 43 22 21 16 15 50 51 0 0 23 0 17 14 0
   56 55 41 24 19 0 13 58 59 0 0 25 26 27 12 0 62 0 39 30 0 28
   9 10 63 64 38 31 32 33 8 5 0 1 37 0 35 34 0 6 3 2
5. 8
   19 0 21 22 25 26 27 28 18 39 38 0 24 0 32 29 0 0 37 36 0 34
   31 0 16 41 42 43 46 47 0 49 15 8 0 44 45 58 57 0 14 9 6 1 2
   59 56 51 13 10 0 4 3 0 0 52 12 11 64 0 62 61 54 0
6. 9
   65 64 0 62 0 60 0 58 57 0 67 36 37 38 39 40 55 0 69 68 0 34
   0 24 0 54 53 0 31 32 33 26 23 42 43 52 71 30 29 0 27 22 0
   44 0 72 79 80 81 20 21 46 47 50 73 78 0 18 0 4 5 48 49 0 77
   16 1 2 3 6 0 8 75 76 15 0 13 12 11 10 9

TEST OUTPUT:

1. 48 46 17 19 21 2 12 25 32 39 4 30 7 37
2. 5 40 47 17 37 12 30 25 27
3. 2 42 35 39 6 44 25 47 10 54 27 16 18 59
4. 44 46 49 54 42 20 57 18 60 40 11 61 29 4 36 7
5. 20 23 33 17 40 35 30 48 7 50 5 60 55 63 53
6. 63 61 59 66 56 35 25 41 70 28 45 51 17 19 74 7 14

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问题: *Numbrix* 是一个n x n 的网格数字谜题。目标是将从1 到 n^2 的所有整数填入网格，即把所有缺失的整数 x 放入网格中，使得 $x - 1$ 和 $x + 1$ 分别与 x 在同一行或同一列相邻。

在本题中，已知网格的大小 n，以及按行优先顺序填充网格的整数 1 到 n^2。所有缺失的数字用 0 表示。你的任务是确定缺失的数字，并按行优先顺序输出它们。

我们需确保：

- 这个谜题有且仅有唯一解。
- 数字 1 和 n^2 绝不会是缺失数字。
- 缺失数字的数量不超过 $2n$ 。

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示例：

输入	5 3 0 1 0 23 0 11 0 25 22 5 10 13 20 0 6 0 14 19 18 0 8 0 16 0
输出	2 24 4 12 21 9 7 15 17
解析	该谜题在二维数组中的表示方式如下： 3 0 1 0 23 0 11 0 25 22 5 10 13 20 0 6 0 14 19 18 0 8 0 16 0 为使该谜题成立，所有缺失的位置必须填入以下整数： 2, 4, 7, 9, 12, 15, 17, 21, 24. 3 2 1 24 23 4 11 12 25 22 5 10 13 20 21 6 9 14 19 18 7 8 15 16 17 **按行优先顺序输出这些整数**: 2 24 4 12 21 9 7 15 17

输入: 输入一个整数 n 表示网格的尺寸，以及一个字符串，该字符串中包含 n x n 二维数组中的 1 到 n^2 所有整数，缺失的数字用 0 表示。n 的取值范围是 4 到 9（包含两端）。

输出: 输出一个由整数组成的字符串，表示按行优先顺序排列的缺失数字，每个数字之间用单个空格分隔。

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样例输入：

1. 5
   3 0 1 0 23 0 11 0 25 22 5 10 13 20 0 6 0 14 19 18 0 8 0 16 0
2. 4
   7 8 0 0 6 1 12 11 5 0 0 0 0 3 16 15
3. 6
   28 0 26 23 0 21 0 30 0 24 19 20 0 31 36 1 0 17 33 0 35 0 15 16 0 5 0 3 14 13 7 8 0 10 11 0
4. 5
   11 0 25 24 23 12 9 8 0 0 0 0 1 6 21 14 3 4 5 0 0 16 0 18 0
5. 6
   27 28 0 30 31 32 0 25 0 17 0 0 1 24 0 16 0 36 2 23 20 15 14 13 3 0 21 8 9 12 4 5 6 0 10 11
6. 6
   16 0 18 0 26 0 15 0 19 24 29 28 14 21 22 0 0 31 0 2 3 4 5 32 12 1 0 7 6 33 0 10 9 36 0 34

样例输出：

1. 2 24 4 12 21 9 7 15 17
2. 9 10 2 13 14 4
3. 27 22 29 25 32 18 34 2 6 4 9 12
4. 10 7 22 13 2 20 15 17 19
5. 29 26 18 34 33 19 35 22 7
6. 17 25 27 20 23 30 13 8 11 35

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测试数据

测试输入:

1. 7
   49 0 47 0 45 44 43 16 0 18 0 20 0 42 15 14 13 24 23 22 41 0
   1 0 0 0 33 40 3 10 11 26 31 34 0 0 9 8 27 0 35 38 5 6 0 28
   29 36 0
2. 7
   3 2 1 42 43 44 49 4 0 6 41 0 45 48 9 8 7 38 39 46 0 10 0 18
   0 36 35 34 11 16 19 20 21 32 33 0 15 24 23 22 31 0 13 14 0
   26 0 28 29
3. 8
   3 0 33 34 37 38 41 0 4 1 32 0 36 0 40 43 5 0 31 24 23 46 45
   0 8 7 30 0 22 0 52 53 9 0 29 26 21 48 51 0 12 11 28 0 20 49
   50 55 13 0 17 0 19 60 0 56 14 15 64 63 62 61 58 57
4. 8
   0 45 0 47 48 0 52 53 43 22 21 16 15 50 51 0 0 23 0 17 14 0
   56 55 41 24 19 0 13 58 59 0 0 25 26 27 12 0 62 0 39 30 0 28
   9 10 63 64 38 31 32 33 8 5 0 1 37 0 35 34 0 6 3 2
5. 8
   19 0 21 22 25 26 27 28 18 39 38 0 24 0 32 29 0 0 37 36 0 34
   31 0 16 41 42 43 46 47 0 49 15 8 0 44 45 58 57 0 14 9 6 1 2
   59 56 51 13 10 0 4 3 0 0 52 12 11 64 0 62 61 54 0
6. 9
   65 64 0 62 0 60 0 58 57 0 67 36 37 38 39 40 55 0 69 68 0 34
   0 24 0 54 53 0 31 32 33 26 23 42 43 52 71 30 29 0 27 22 0
   44 0 72 79 80 81 20 21 46 47 50 73 78 0 18 0 4 5 48 49 0 77
   16 1 2 3 6 0 8 75 76 15 0 13 12 11 10 9

测试输出：

1. 48 46 17 19 21 2 12 25 32 39 4 30 7 37
2. 5 40 47 17 37 12 30 25 27
3. 2 42 35 39 6 44 25 47 10 54 27 16 18 59
4. 44 46 49 54 42 20 57 18 60 40 11 61 29 4 36 7
5. 20 23 33 17 40 35 30 48 7 50 5 60 55 63 53
6. 63 61 59 66 56 35 25 41 70 28 45 51 17 19 74 7 14